

Pav Bhaji Text Classification

Machine Learning and NLP Project Report

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Project Type: Text Classification using Machine Learning

Domain: Natural Language Processing / Food Content Classification

This project focuses on building a machine learning system that can identify whether an Instagram food post is related to Pav Bhaji based on its caption and hashtags.

1. Introduction

Social media food posts contain useful textual information such as captions, hashtags, mentions, and descriptions. This project uses that text to solve a binary classification problem: determining whether a food post is about Pav Bhaji or not.

The project follows an end-to-end machine learning workflow. The raw Instagram data was first inspected and processed, captions and tags were combined, unnecessary text was removed, and the resulting text was converted into numerical features using TF-IDF. Multiple classification algorithms were then trained and evaluated.

2. Problem Statement

To develop a text classification model that can automatically classify Instagram food-post text into two classes: **Pav Bhaji** and **Not Pav Bhaji**.

3. Objectives

- Build a complete NLP preprocessing pipeline for social-media food text.
- Combine captions and hashtags into a usable text field.
- Clean URLs, mentions, hashtags, newline characters, numbers, and extra spaces.
- Convert cleaned text into TF-IDF features.
- Train and compare Logistic Regression, Linear SVM, and Naive Bayes models.
- Evaluate the models using accuracy, precision, recall, F1 score, and a confusion matrix.
- Test the trained classifier on new text inputs.
- Save the trained model and processed analysis outputs for reuse.

4. Dataset Description

The project dataset contains Instagram food-post information including captions, tags, and image-related metadata. After processing the available records, the machine-learning dataset contained 452 labelled examples.

Class	Label	Records
Non-Pav Bhaji	0	269
Pav Bhaji	1	183
Total		452

5. Data Preprocessing

The raw caption and tag information contained social-media noise. A preprocessing function was used to normalize the text before model training.

- Convert text to lowercase.
- Remove URLs and web links.

- Remove user mentions beginning with @.
- Remove hashtag symbols while retaining the associated words.
- Normalize newline characters.
- Remove non-word characters and numbers.
- Remove repeated whitespace and trim the final text.

6. Feature Engineering

After cleaning, the caption and tag information was represented as text features. TF-IDF (Term Frequency–Inverse Document Frequency) was used to convert the text into a numerical representation suitable for machine learning algorithms.

TF-IDF gives higher importance to terms that are useful for distinguishing documents while reducing the influence of words that occur very frequently across the dataset.

7. Model Development

Three supervised machine learning algorithms were evaluated for the binary classification task.

Model	Purpose
Logistic Regression	Linear probabilistic classifier for binary text classification
Linear SVM	Margin-based linear classifier commonly used for high-dimensional text data
Naive Bayes	Probabilistic classifier based on conditional independence assumptions

8. Model Evaluation

The models were compared using accuracy, precision, recall, and F1 score. The recorded results from the project notebook are shown below.

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	0.6264	0.5254	0.8378	0.6468
Linear SVM	0.6264	0.5319	0.6757	0.5952
Naive Bayes	0.5604	0.4571	0.4324	0.4444

9. Confusion Matrix

A confusion matrix was generated for the Logistic Regression classifier. The matrix recorded the following counts, with rows representing actual classes and columns representing predicted classes:

Actual / Predicted	Non-Pav Bhaji	Pav Bhaji
Non-Pav Bhaji	26	28
Pav Bhaji	6	31

The matrix shows 26 correctly classified Non-Pav Bhaji posts, 31 correctly classified Pav Bhaji posts, 28 Non-Pav Bhaji posts classified as Pav Bhaji, and 6 Pav Bhaji posts classified as Non-Pav Bhaji.

10. Keyword-Removal Experiment

An additional experiment was performed by removing the explicit Pav Bhaji keyword before classification. The Logistic Regression model in this experiment produced an accuracy of 0.6044, precision of 0.5088, recall of 0.7838, and F1 score of 0.6170.

This experiment was used to examine how much the classifier depends on direct Pav Bhaji keywords in the social-media text.

11. Prediction Testing

The saved classifier was tested with new text inputs to verify that the complete inference pipeline could clean the input, apply the trained model, and return a class label.

Input	Prediction
Delicious Mumbai street food with spicy vegetables and buttered pav	Pav Bhaji
Fresh chicken tikka with salad and garlic naan for dinner	Not Pav Bhaji

12. Project Structure

```
drivebuddy-pavbhaji-ml/  
■■■ data/dataset/images/  
■■■ data/dataset/pavbhaji.json  
■■■ models/pavbhaji_text_classifier.pkl  
■■■ notebooks/pavbhaji_classification.ipynb  
■■■ reports/model_comparison.csv  
■■■ reports/processed_dataset.csv  
■■■ reports/misclassified_samples.csv  
■■■ src/pavbhaji_classifier.py  
■■■ requirements.txt  
■■■ README.md
```

13. Technologies Used

- Python
- Pandas and NumPy
- Scikit-learn
- Joblib
- Matplotlib and Seaborn
- Jupyter Notebook

14. Limitations

The dataset is relatively small and the two classes are not perfectly balanced. Social-media text also contains informal language, spelling variations, emojis, hashtags, and promotional content, which can make classification difficult. The reported results therefore represent this particular dataset and experimental setup rather than a general measure of performance on all Instagram food posts.

15. Future Scope

- Collect a larger and more diverse dataset.
- Improve handling of slang, emojis, spelling variations, and multilingual text.
- Experiment with additional NLP and machine learning approaches.
- Combine text-based features with image-based features.
- Build a web interface for real-time predictions.
- Evaluate the system on a separate external test dataset.

16. Conclusion

This project demonstrates an end-to-end NLP classification workflow for identifying Pav Bhaji-related food content from Instagram text. The work covers data preparation, text cleaning, TF-IDF feature extraction, model training, evaluation, error analysis, and deployment-oriented model saving.

The project also provided practical experience with comparing classification algorithms and testing a trained model on previously unseen text.

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